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Major Project Presentation

LEEN AI

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Introduction

- Traditional education follows a **one-size-fits-all** teaching approach, which does not meet the individual learning needs of every student.
- Students have different learning abilities, understanding levels, and learning speeds, requiring a more personalized approach.
- Teachers often cannot provide individual attention to every student due to large class sizes and limited classroom time.
- Students frequently experience delays in doubt resolution outside classroom hours, affecting their learning continuity.
- Existing learning platforms mainly provide static content and lack personalized guidance, adaptive learning, and continuous performance analysis.
- **LEEN AI – An AI Personal Teacher System** is designed to overcome these challenges by providing personalized learning, intelligent doubt solving, adaptive study planning, automated assessments, and real-time performance tracking.
- The system uses **Artificial Intelligence (AI), Natural Language Processing (NLP),** to deliver accurate, context-aware, and interactive learning support.
- It is available **24×7**, enabling students to learn anytime and anywhere while promoting self-paced, flexible, and student-centric education.

Identification of Problem Domain

1. Lack of Personalized Learning
Every student receives the same teaching method regardless of their understanding level. Fast learners become bored, while slow learners struggle to keep up.

2. Limited Teacher Availability
Teachers cannot be available all the time. Students often have doubts while studying at home but cannot get immediate help.

3. Poor Performance Tracking
Most schools evaluate students only through exams and marks. This does not accurately represent a student's understanding or learning progress.

4. Static Learning Materials
Books, notes, and videos are fixed and cannot adapt according to individual student needs.

5. Lack of Proper Study Planning
Many students do not know what to study, when to revise, or how to prepare effectively.

Feasibility of Project Proposal

Technical Feasibility

✓ Built using modern technologies such as **React.js, FastAPI, MongoDB, Redis, Qdrant, NLP, and RAG**.

✓ Supports AI-driven personalization, semantic search, and real-time analytics.

✓ Modular architecture ensures scalability, security, and easy maintenance.

Operational Feasibility

✓ Simple and intuitive interface for **Students, Parents, and Admin**.

✓ Available **24×7** for continuous learning and instant doubt solving.

✓ Reduces teachers' workload while improving student engagement and learning outcomes.

Economic Feasibility

✓ Developed using **open-source technologies**, minimizing software costs.

✓ Low maintenance cost with scalable deployment using Docker.

✓ Cost-effective solution compared to private tutoring and commercial learning platforms.

Schedule Feasibility

✓ Follows a structured Software Development Life Cycle (SDLC).

✓ Can be completed within the academic project duration through planned phases.

✓ Development includes Requirement Analysis, Design, Implementation, Testing, and Deployment.

Objectives

The primary objective of LEEN AI is to develop an intelligent educational platform capable of providing personalized and adaptive learning experiences.

- Develop an AI-powered personalized learning system.
- Provide instant doubt-solving through text, voice, and image inputs.
- Generate adaptive study schedules according to individual learning progress.
- Automatically generate quizzes and evaluate student performance.
- Continuously monitor academic progress using AI analytics.
- Identify weak learning areas and recommend targeted improvements.
- Increase student engagement through interactive learning techniques.
- Enable parents to monitor their child's academic performance through dashboards and reports.
- Improve learning efficiency using Retrieval-Augmented Generation (RAG) and Natural Language Processing.

Methodology of the Proposed Work

The proposed system is developed using an iterative and incremental (Agile) model, in which every module is designed, implemented, tested and refined before the next phase is taken up.

- **Phase 1 – Requirement Analysis:** Study of existing e-learning platforms and their limitations, and collection of requirements from students, teachers and parents.
- **Phase 2 – System Design:** Design of the three-tier architecture, the database (ER) schema, versioned REST API contracts, data-flow diagrams and user-interface wireframes.
- **Phase 3 – Knowledge Base Preparation:** Collection of syllabus-aligned material, text extraction and cleaning, semantic chunking, and storage of the resulting vector embeddings.
- **Phase 4 – AI Module Development:** Implementation of the Retrieval-Augmented Generation (RAG) pipeline, NLP-based doubt solving, automatic quiz generation and the analytics engine.
- **Phase 5 – Application Development:** Development of the React frontend and FastAPI backend with PostgreSQL, Redis and Celery, integrated through secure, versioned REST APIs.
- **Phase 6 – Testing and Deployment:** Unit, integration and user acceptance testing, containerised deployment using Docker and Nginx, and refinement based on user feedback.

Methodology – System Architecture

The system is organised as a modular three-tier architecture in which the presentation, application and data layers are loosely coupled and communicate only through versioned REST APIs:

- **Presentation Layer:** A React (Vite) and TailwindCSS single-page application that provides separate dashboards for students, parents and teachers.
- **Application Layer:** FastAPI (Python) services that handle authentication, request validation and business logic, with Celery and Redis executing long-running tasks in the background.
- **Intelligence (AI) Layer:** Embedding generation, semantic retrieval from the knowledge base and context-grounded response generation by the language model using the RAG technique.
- **Data Layer:** PostgreSQL for structured data such as users, quizzes and progress records, Redis for caching and task queuing, and a vector database (Qdrant) for the semantic knowledge base.
- **Deployment Layer:** All services are containerised with Docker, orchestrated using Docker Compose and exposed through an Nginx reverse proxy.

Functional Modules: Doubt Solving, Adaptive Study Planner, Quiz and Assessment, Performance Analytics, Parent Monitoring, and Authentication and Profile Management.

Methodology – Working Flow

The working of the proposed system follows the sequence given below:

- **Step 1:** The student registers and logs in; the profile, class and initial learning level are stored securely in the database.
- **Step 2:** The student submits a doubt as text, voice or an image, and the input is pre-processed and normalised using NLP techniques.
- **Step 3:** The processed query is converted into a vector embedding and semantically matched against the syllabus knowledge base.
- **Step 4:** The most relevant content chunks are retrieved and passed, together with the query, to the language model (Retrieval-Augmented Generation).
- **Step 5:** A context-grounded, syllabus-aligned explanation is generated and returned to the student in real time.
- **Step 6:** Every interaction, quiz attempt and score is recorded to build a continuous learning profile of the student.
- **Step 7:** The analytics engine evaluates this profile to identify weak topics and to measure conceptual progress over time.
- **Step 8:** The adaptive planner regenerates the study schedule and revision plan, and the results are rendered on the student and parent dashboards.

Report generation and notifications are executed asynchronously through Celery workers, so that the response time of the learning interface remains unaffected.

Conclusion

LEEN AI represents a next-generation intelligent educational platform that combines Artificial Intelligence with personalized learning methodologies.

The proposed system overcomes several limitations of traditional education by providing:

- Personalized concept explanation
- Adaptive study planning
- Automated assessments
- Continuous performance analytics
- Parent monitoring dashboards
- Instant AI-based doubt solving

Although the current version has limitations such as internet dependency and AI accuracy, future enhancements may include multilingual support, speech-based interaction, mobile applications, predictive learning analytics, and integration with educational institutions.

In conclusion, LEEN AI has the potential to revolutionize digital education by making learning more intelligent, adaptive, interactive, and accessible for students worldwide.

References

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